# Responses to Reviewers and Cover letter

**Manuscript ID:** IPM-D-25-06425

**Manuscript Title:** ALD$^2$: Adaptive Layer-wise Denoising Decoding for Hallucinations Mitigation in Large Vision-Language Models

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Dear Editor-in-Chief,

We would like to express our sincere appreciation to all the reviewers and editors for their valuable time and effort given to reviewing our paper. Their constructive comments will greatly improve the quality of our paper. According to the suggestions of all reviewers, we have made corresponding revisions as detailed below.

**Editor-in-Chief (comments and responses):**

Thank you for submitting your manuscript to Information Processing and Management.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following minor revision and modification. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by 03/29/2026.

When revising your manuscript, please carefully consider all issues mentioned in the reviewers' comments: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

To submit your revised manuscript, please log in as an author at https://www.editorialmanager.com/ipm/, and navigate to the "Submissions Needing Revision" folder under the Author Main Menu.

Information Processing and Management values your contribution and I look forward to receiving your revised manuscript.

**Response:** Thank you for your response and for coordinating the evaluation of our manuscript. We would like to express our gratitude for the opportunity to revise and resubmit our manuscript following the major revision recommendation. We have been dedicated to ensuring that all concerns raised by the reviewers are effectively addressed in the revised version of the manuscript. We have made every effort to provide detailed responses and implement necessary modifications to meet the expectations of the reviewers and improve the overall quality of the paper.

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**Associate Editor (comments and responses)**

Thank you for submitting your paper to IP&M. We have reviews on your manuscript, and the reviewers are supportive of your research but also have some important points, including the issues raised by the reviewers below, that I think would be useful to clarify in a revision of your article.

Depending on your revisions and response to the reviewers, I will determine whether or not to send it out for another round of review. If the revisions are satisfactory, another round of review might not be necessary.

Therefore, I invite you to respond to the comments and revise your manuscript. If you decide to resubmit, please include a detailed response to the reviewer highlighting your changes. Given the high volume of submissions to IP&M and the desire to maintain the journal’s rapid processing time, a detailed response to the reviewers aids significantly in this regard. To assist with this, here is one of many nice articles on crafting a detailed response to the reviewers - https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005730#sec006.

**Response:** We sincerely thank you for your thoughtful feedback and for coordinating the evaluation of our manuscript. We greatly appreciate the opportunity to revise and resubmit our work. In the revised paper, we have addressed all the reviewers' concerns. The specific revisions are as follows:

① We add the relevant citations suggested by reviewer in Related work section.

# To All Reviewers:

We sincerely thank you for your detailed and professional review comments, which have provided valuable guidance for further improving the quality of our paper.
In response to your suggestions and questions, we have made corresponding revisions and refined the content of the manuscript accordingly. The following presents the specific modifications we have made:

**Attention:**
In the following document, black text denotes the your comments, blue text presents our responses, gray text highlights the relevant context from the manuscript, and red text indicates the corresponding revisions of the manuscript.

## To Reviewer #1:

The authors have carefully addressed all of my previous comments and concerns in the revised manuscript. The revisions are clear, thorough, and adequately justified. In particular, the additional explanations and experimental results significantly improve the clarity and completeness of the paper. I have no further questions or concerns at this stage. The manuscript is now suitable for publication, and I recommend acceptance.

We sincerely thank you for the careful evaluation of our revised manuscript and for the positive feedback. We are glad that the revisions, additional explanations, and experimental results have addressed the previous concerns and improved the clarity and completeness of the paper. We greatly appreciate your recognition of our efforts and the recommendation for acceptance.

## To Reviewer #2:

**Comment #2.1:**

Although the authors stated that they have expanded the related work to cover the recent studies I suggested (including grounding/faithfulness and decoding-time mitigation), I could not find the corresponding additions in the revised paper. The previous comment is: The related work on hallucination mitigation appears incomplete. I recommend expanding the discussion to include more recent studies on hallucinations in LVLMs, particularly works focusing on grounding, faithfulness, and decoding-time mitigation. For example, FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models, Aligning large multimodal models with factually augmented rlhf, A Unified Hallucination Mitigation Framework for Large Vision-Language Models. Mmbench: Is your multi-modal model an all-around player? Fine-grained and Explanable Factuality Evaluation for Multimodal Summarization. FIFA: Unified Faithfulness Evaluation Framework for Text-toVideo and Video-to-Text Generation, Decide: Alleviating Hallucination in Large Vision-Language Models via Multi-View Multi-Path Reasoning, FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback. From pixels to tokens: Revisiting object hallucinations in large vision-language models

We thank you for the careful reading and for pointing out this issue. In the previous revision, we expanded the discussion in the Related work section to cover studies on grounding, faithfulness, and decoding-time mitigation for hallucinations in LVLMs. ***However, we regret that the corresponding citations to the recommended works were inadvertently omitted in the reference list during the revision process.***

In the latest revised version, we have corrected this oversight by explicitly adding the suggested references, including:

1. FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models,
2. Aligning Large Multimodal Models with Factually Augmented RLHF,
3. A Unified Hallucination Mitigation Framework for Large Vision-Language Models,
4. MmBench: Is your Multi-modal Model an All-around Player?,
5. Fine-grained and Explainable Factuality Evaluation for Multimodal Summarization,
6. FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation,
7. Look, compare, decide: Alleviating Hallucination in Large Vision-Language Models via Multi-View Multi-Path Reasoning,
8. FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback,
9. From Pixels to Tokens: Revisiting Object Hallucinations in Large Vision-Language Models.

These works are now properly cited and discussed in the Related Work section. We appreciate the reviewer's helpful suggestion, which improves the completeness of the related work discussion. The relevant revised contexts are as follows:

5. Related work

5.1 Hallucinations Mitigation in LVLMs

Hallucination mitigation in LVLMs has attracted increasing attention due to the inherent difficulty of aligning visual and textual modalities. Prior studies mainly explore three directions (Huo et al., 2024). The first direction strengthens model robustness through instruction tuning or alignment with curated, visually grounded datasets (Lee et al., 2022; Gunjal et al., 2024; Liu et al., 2023; Jing et al., 2024; 2025; Liu et al., 2024c; Zhang et al., 2024), including approaches based on factually augmented reinforcement learning and fine-grained feedback to improve visual faithfulness (Jing & Du, 2024; Sun et al., 2024; Shang et al., 2025b). The second direction applies post-hoc correction or verification with auxiliary analysis modules to detect and suppress hallucinated content after generation (Manakul et al., 2023; Zhou et al., 2024a; Chang et al., 2024), often relying on external reasoning paths, multi-view analysis, or additional verification signals. The third line of work adjusts the decoding process to mitigate hallucinations through contrastive, constraint-based, or uncertainty-aware strategies (Li et al., 2023b; Chuang et al., 2023; Liu et al.,2024b). Representative methods include visual contrastive decoding and its variants (e.g., VCD and Med-VCD) (Mahdavi et al., 2026; Leng et al., 2024), instruction contrastive decoding (ICD) (Wang et al., 2024b), dynamic correction decoding (DeCo) (Wang et al., 2024a), multi-view multi-path reasoning (Qu et al., 2025), and self-introspective decoding (SID) (Huo et al., 2024), which dynamically evaluates token importance during generation. These decoding-time approaches are lightweight and model-agnostic and have shown promising effectiveness in reducing hallucinations without retraining large models.

Qu, X., Sun, J., Wei, W., Liu, D., Dong, J., & Cheng, Y. (2025). Look, compare, decide: Alleviating hallucination in large vision-language models via multi-view multi-path reasoning. In Proceedings of the 31st International Conference on Computational Linguistics (pp. 4428–4441).

Jing, L., & Du, X. (2024). Fgaif: Aligning large vision-language models with fine-grained ai feedback. Transactions on Machine Learning Research.

Jing, L., Lai, V. D., Yoon, S., Bui, T., & Du, X. (2025). Fifa: Unified faithfulness evaluation framework for text-to-video and video-to-text generation. In Knowledgeable Foundation Models at ACL 2025.

Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z. et al. (2024c). Mmbench: Is your multi-modal model an all-around player? In European conference on computer vision (pp. 216–233). Springer.

Shang, Y., Zeng, X., Zhu, Y., Yang, X., Fang, Z., Zhang, J., Chen, J., Liu, Z., & Tian, Y. (2025b). From pixels to tokens: Revisiting object hallucinations in large vision-language models. In Proceedings of the 33rd ACM International Conference on Multimedia (pp. 10496–10505).

Chang, Y., Jing, L., Zhang, X., & Zhang, Y. (2024). A unified hallucination mitigation framework for large vision-language models. Transactions on Machine Learning Research.

Zhang, Y., Zuo, J., Su, K., & Jing, L. (2024). Fine-grained and explainable factuality evaluation for multimodal summarization. arXiv preprint arXiv:2402.11414.

Sun, Z., Shen, S., Cao, S., Liu, H., Li, C., Shen, Y., Gan, C., Gui, L., Wang, Y.-X., Yang, Y. et al. (2024). Aligning large multimodal models with factually augmented rlhf. In Findings of the Association for Computational Linguistics: ACL 2024 (pp. 13088–13110).

Jing, L., Li, R., Chen, Y., & Du, X. (2024). Faithscore: Fine-grained evaluations of hallucinations in large vision-language models. In Findings of the Association for Computational Linguistics: EMNLP 2024 (pp. 5042–5063).